Schur Rings Over Numerical Semigroups

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Numerical Semigroups

Definition

A Numerical Semigroup is a subset of the natural numbers which is closed under addition.

Example

- $\mathbb{Z}^+$
- $n\mathbb{Z}^+, \forall n \in \mathbb{Z}^+$
- $\{3, 5, 6, 8, 9, 10, 11, \rightarrow\}$

Properties of Numerical Semigroups

Any numerical semigroup can be reduced to a finite generating set.

Example

- $\mathbb{Z}^+ = \langle 1 \rangle$
- $n\mathbb{Z}^+ = \langle n \rangle$
- $\{3, 5, 6, 8, 9, 10, 11, \rightarrow\} = \langle 3, 5 \rangle$

It can be assumed the gcd of a generating set is 1.

Properties of Numerical Semigroups, Cont.

A numerical semigroup with at least two generators greater than 0 has a Frobenius number and a conductor.

Definition

The Frobenius number is the greatest integer not in the semigroup, e.g. $F(\langle 3, 5\rangle) = 7$

Definition

The Conductor is the smallest element of the numerical semigroup such that all greater natural numbers are in the semigroup, e.g. $C(\langle 3, 5\rangle) = F(\langle 3, 5\rangle) + 1 = 8$

Notation

Note that $\langle a_0, a_1, ..., a_n \rangle$ can be expressed as $\langle x^{a_0}, x^{a_1}, ..., x^{a_n} \rangle$, as the algebraic structure is identical with polynomial multiplication. This notation is helpful when discussing Schur rings.

Schur Rings

Definition

The coarsest Schur ring is the partition such that if $a, b \in S$ always have the same coefficient in $\mathbb{Z}[S]$, then a, b are combined into the same element of the partition.

Schur Rings Cont.

Let $S = \langle x^3, x^5 \rangle$.
Start by combining the generators, $x^3 + x^5$.
Then check higher orders.

$$(x^3 + x^5)^2 = x^6 + 2x^8 + x^{10}$$

There is no other way to generate x^6 or x^{10}, so they can be combined.
Similarly, there is no other way to generate x^8, so it is a singleton.

Schur Rings Cont.

$$(x^3 + x^5)^3 = x^9 + 3x^{11} + 3x^{13} + x^{15}$$

$x^{15} \in S^5, x^9 \notin S^5$, thus they are both singletons.
x^{11}, x^{13}? They both can only be generated by $x^8(x^3 + x^5)$, so they can be combined.
No further combining is possible.
Final Coarsest Schur Ring: $x^3 + x^5, x^6 + x^{10}, x^8, x^9, x^{11} + x^{13}, x^{12}, \rightarrow$
Finer Schur rings are possible.

Primary Research Question

Conjecture

All Schur Rings over numerical semigroups are eventually discrete.

Theorems

Theorem 1

For $S = \langle p, q \rangle$ where $p < q$ and $\gcd(p, q) = 1$, no element of S greater than $(p+q)(p-1)$ can be combined in a Schur ring.

Theorem 2

For any numerical semigroup of the form $S = \langle m, m+1, m+2, ..., 2m-1 \rangle$, any Schur ring over S is eventually discrete.

Theorem 3

For the numerical semigroup, $S = \langle a_0, a_1, ..., a_k \rangle$, if $\gcd(a_i, a_j) = 1$ for all $0 \leq i, j \leq k$ with $i \neq j$, then any Schur ring over S is eventually discrete.

Sketch of Proof of Theorem 1

Theorem 1

For $S = \langle p, q \rangle$ where $p < q$ and $\gcd(p, q) = 1$, no element of S greater than $(p+q)(p-1)$ can be combined in a Schur ring.

Proof

For $s \in S$, there exists a unique maximal decomposition which contains more generators than all other decompositions.

Two elements of S can only have a maximal decomposition of the same order if they are both less than $(p+q)(p-1)$

A similar theorem for 3 generators was recently developed.

Sketch of Proof of Theorem 2

Theorem 2

For any numerical semigroup of the form $S = \langle m, m+1, m+2, ..., 2m-1 \rangle$, any Schur ring over S is eventually discrete.

Proof

$(x^m + x^{m+1} + ... + x^{2m-1})^2 = x^{2m} + 2x^{2m+1} + ... + bx^{3m-2} + ax^{3m-1} + bx^{3m} + ... + x^{4m-2}$
$(x^m + x^{m+1} + ... + x^{2m-1})^3 = x^{3m} + ...$
Thus $3m-1$ and $3m$ are both singletons. Note also $\gcd(3m-1, 3m) = 1$. This generates a numerical semigroup within the Schur ring, which is eventually discrete.

Sketch of Proof of Theorem 3

Theorem 3

For a numerical semigroup, $S = \langle a_0, a_1, ..., a_k \rangle$, if $\gcd(a_i, a_j) = 1$ for all $0 \leq i, j \leq k$ with $i \neq j$, then any Schur ring over S is eventually discrete.

Proof

The case for 2 and 3 generators are covered by other theorems. For 4 or more, two coprime singletons can be guaranteed within the Schur ring.

New Paradigm

Definition

"Freshman Exponentiation" for a polynomial $X = a_0 + a_1x + \ldots + a_nx^n$, denoted $X^{(m)}$ is defined as $X^{(m)} = a_0^m + (a_1x)^m + \ldots + (a_nx^n)^m$.

Definition

The Hadamard product of two polynomials $X_a = a_0 + a_1x + \ldots + a_nx^n, X_b = b_0 + b_1x + \ldots + b_mx^m$, denoted $X_a \circ X_b$ is defined as $X_a \circ X_b = \sum_{i=0}^{k} a_ib_ix^i$ where $k = \max(n, m)$.

A Schur ring is closed under both operations.

Two Generator Case

Let $S = \langle a, b \rangle$. Thus $x^a + x^b$ is an element of the Schur Ring over S.
Consider $(x^a + x^b)^2 = x^{2a} + 2x^{a+b} + x^{2b}$. No two terms of this polynomial can be combined, as $a \neq b$, thus x^{a+b} is a singleton in the partition for the Schur Ring.
Consider $(x^a + x^b)^{(a)} \circ (x^a + x^b)^{(b)}$.

$$\begin{aligned}
&(x^a + x^b)^{(a)} \circ (x^a + x^b)^{(b)} \\
&= (x^{a^2} + x^{ab}) \circ (x^{ab} + x^{b^2}) \\
&= x^{ab} \text{ because } a \neq b
\end{aligned}$$

Thus x^{ab} is a singleton in the Schur Ring.

Two Generator Case Cont.

Note $\gcd(a+b, ab) = 1$, as $\gcd(a, b) = 1$.
Therefore x^{a+b}, x^{ab} form a numerical semigroup in the Schur Ring, meaning the Schur Ring is eventually discrete.
Note the argument can be extended to 3 and 4 generators.
A modified form of this argument can be inducted to any number of generators.

References

Abdallah Assi, Pedro A. Garcia-Sanchez
Numerical Semigroups and Applications

The End